

School of Behavioral and Brain Sciences

Neuroscience (BS)

Neuroscience is the multidisciplinary study of nervous system function that draws on recent advances in cell and molecular biology, biochemistry, biophysics, computer, behavioral and cognitive sciences. It examines the brain, spinal cord, and peripheral nervous system's global and nanoscale biochemistry, its complex and extensively networked morphology, and its remarkably adaptive physiology. The field considers neuronal development from early embryology through advanced senescence. Further, neurobiologists examine neuronal plasticity from the level of single proteins, of individual neurons, up through the level of networks or systems of cells, and up to complete behaving organisms. Neuroscience studies the regulation and expression of behavior, the impact of that behavior on neurons, and the complex interactions of multiple neuronal systems that underlie the emergence of cognitive and sensory functions. The Neuroscience program at UT Dallas provides students with opportunities to focus on the brain, spinal cord, and peripheral neurons from a systems and cellular-level perspective, drawing on behavioral and cognitive expertise combined with smaller-scale neuronal and molecular analyses. It allows undergraduates extensive interactions with working neuroscientists who use the latest experimental techniques.

The Neuroscience program is designed to prepare students for admission to medical, dental or allied-health graduate schools (through the Medical Neuroscience track), for admission to research-based doctoral training programs (through the Research Neuroscience track), or for careers in biomedical science, industry, and allied health fields (through the Industrial Neuroscience track). Required courses within the Medical track include the approved pre-medical curriculum and offer a popular alternative to other pre-health majors. Students who wish to continue their education in the fields of medicine, dentistry or allied health professions should also register with the Health Professions Advising Center during their first semester.

Students can choose between three program tracks: Medical Neuroscience Track, Research Neuroscience Track, or Industrial Neuroscience Track to meet their career training needs. All tracks share a Core Curriculum and Neuroscience major requirements that can be completed in a minimum of 82 semester credit hours. Different coursework is required within the different tracks, with the total number of credit hours required for graduation the same for all three tracks.

Bachelor of Science in Neuroscience

[Degree Requirements](#) (120 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

Faculty

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Professors: Gregory Dussor, Michael P. Kilgard, Theodore Price, Robert L. Rennaker II, Steven Small, Ana Solodkin

Associate Professors: Michael Burton, Benedict Kolber, Sven Kroener, Christa McIntyre Rodriguez

Assistant Professors: Andrew Eagle, Crystal Engineer, Zirong Gu, Lena Nguyen, Puja Parekh, Millie Rincon-Cortes, Katelyn Sadler, Diana Tavares Ferreira, Catherine Thorn

Professor of Instruction: Van Miller

Associate Professor of Instruction: Steven McWilliams, Siham Raboune, Rukhsana Sultana, Anna Taylor

Assistant Professors of Instruction: Rafael Granja-Vasquez, Faisal Jahangiri, William Marks, Cecily Oleksiak, Amy Zwierzchowski-Zarate

I. Core Curriculum Requirements: 42 semester credit hours¹

Communication: 6 semester credit hours

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 2414](#) Integral Calculus²

or [MATH 2417](#) Calculus I²

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

Life and Physical Sciences: 6 semester credit hours

[BIOL 2311](#) Introduction to Modern Biology I²

[CHEM 1311](#) General Chemistry I²

Or select 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see NSC advisor)

Language, Philosophy and Culture (040): 3 semester credit hours

Select 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

Select 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

[PSY 2301](#) Introduction to Psychology²

Or select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

[PSY 2317](#) Statistics for Psychology²

or [STAT 1342](#) Statistical Decision Making²

or [STAT 2332](#) Introductory Statistics for Life Sciences²

[CHEM 1312](#) General Chemistry II²

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 26 semester credit hours

Major Preparatory Courses: 24 semester credit hours (5 semester credit hours beyond Core Curriculum)²

All of the following:

[BIOL 2111](#) Introduction to Modern Biology Workshop I

[BIOL 2281](#) Introductory Biology Laboratory

[BIOL 2311](#) Introduction to Modern Biology I²

[CHEM 1111](#) General Chemistry Laboratory I

[CHEM 1311](#) General Chemistry I₂

[CHEM 1112](#) General Chemistry Laboratory II

[CHEM 1312](#) General Chemistry II₂

[MATH 2414](#) Integral Calculus₂

or [MATH 2417](#) Calculus I₂

[PSY 2301](#) Introduction to Psychology₂

[PSY 2317](#) Statistics for Psychology₂

or [STAT 1342](#) Statistical Decision Making₂

or [STAT 2332](#) Introductory Statistics for Life Sciences₂

Major Core Courses required for all Tracks: 21 semester credit hours

All of the following:

[NSC 3361](#) Introduction to Neuroscience

[NSC 4352](#) Cellular Neuroscience

[NSC 4353](#) Neuroscience Laboratory Methods

[NSC 4354](#) Integrative Neuroscience

[NSC 4356](#) Neurophysiology

[NSC 4363](#) Neuropharmacology

[NSC 4366](#) Neuroanatomy

III. Tracks - 52 semester credit hours

Neuroscience majors select one of three Tracks:

- [Medical Neuroscience](#) (12 semester credit hours, plus 28 semester credit hours of pre-med courses, plus 12 semester credit hours of free electives)
- [Research Neuroscience](#) (12 semester credit hours, plus 28 semester credit hours of pre-graduate courses, plus 12 semester credit hours of free electives)
- [Industrial Neuroscience](#) (22 semester credit hours, plus 30 semester credit hours of free electives)

Track 1: Medical Neuroscience

Choose four courses (12 semester credit hours) from the following:

[NSC 4310](#) Principles of Pediatric Neurology

[NSC 4320](#) Sleep and Sleep Disorders

[NSC 4330](#) Fundamentals of EEG and ECoG

[NSC 4350](#) Medical Neuropathology

[NSC 4351](#) Medical Neuroscience

[NSC 4358](#) Neuroscience of Pain

[NSC 4362](#) Molecular Neuroscience

[NSC 4364](#) Journey into Medicine

[NSC 4372](#) Neuroimmunology

[NSC 4373](#) Sensory Neuroscience

[NSC 4377](#) Neurogenetics

[NSC 4378](#) Neurotoxicology

[NSC 4380](#) Neurology of Cognition

[NSC 4381](#) Health Disparities in Neuroscience

[NSC 4382](#) Neurobiology of Emotion

[NSC 4383](#) Human Neurophysiology Lab

[NSC 4387](#) Neuropathology

[NSC 4388](#) Medical Physiology

[NSC 4397](#) Thesis Research

[NSC 4V75](#) Honors Seminar

[NSC 4V98](#) Directed Research³

[NSC 4V99](#) Independent Study⁴

Required Pre-medical Basic Biology, Chemistry and Physics (28 semester credit hours)

[BIOL 2112](#) Introduction to Modern Biology Workshop II

[BIOL 2312](#) Introduction to Modern Biology II

[BIOL 3461](#) Biochemistry I

[CHEM 2233](#) Introductory Organic Chemistry Laboratory

[CHEM 2323](#) Introductory Organic Chemistry I

[CHEM 2325](#) Introductory Organic Chemistry II

[PHYS 1301](#) College Physics I

[PHYS 1101](#) College Physics Laboratory I

or [PHYS 2125](#) Physics Laboratory I

[PHYS 1302](#) College Physics II

[PHYS 1102](#) College Physics Laboratory II

or [PHYS 2126](#) Physics Laboratory II

An additional 4 semester credit hours selected with your advisor from NSC or other fields. See advisor for additional guidance.

Elective Requirements: 12 semester credit hours of free electives.

At least 12 semester credit hours of lower- or upper-division courses of the student's choice. Students are encouraged to explore additional courses in Neuroscience as well as explore interests outside the field.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

Incoming freshmen must enroll and complete the requirement of [BBSU 1100](#).

Track 2: Research Neuroscience

Choose four courses (12 semester credit hours) from the following:

[NSC 4320](#) Sleep and Sleep Disorders

[NSC 4330](#) Fundamentals of EEG and ECoG

[NSC 4355](#) Advanced Neuroscience Laboratory

[NSC 4357](#) Neurobiology of Learning and Memory

[NSC 4358](#) Neuroscience of Pain

[NSC 4359](#) Cognitive Neuroscience

[NSC 4362](#) Molecular Neuroscience

[NSC 4367](#) Developmental Neurobiology

[NSC 4371](#) Neural Plasticity

[NSC 4372](#) Neuroimmunology

[NSC 4373](#) Sensory Neuroscience

[NSC 4374](#) Neuroplasticity in Disorders of the Nervous System

[NSC 4376](#) Neurobiology of Stress

[NSC 4377](#) Neurogenetics

[NSC 4380](#) Neurology of Cognition

[NSC 4381](#) Health Disparities in Neuroscience

[NSC 4382](#) Neurobiology of Emotion

[NSC 4383](#) Human Neurophysiology Lab

[NSC 4391](#) Writing and Independent Study

[NSC 4397](#) Thesis Research

[NSC 4V75](#) Honors Seminar

[NSC 4V98](#) Directed Research

[NSC 4V99](#) Independent Study

Required pre-graduate courses, basic Biology, Chemistry and Physics (28 semester credit hours)

[BIOL 2112](#) Introduction to Modern Biology Workshop II

[BIOL 2312](#) Introduction to Modern Biology II

[BIOL 3461](#) Biochemistry I

[CHEM 2233](#) Introductory Organic Chemistry Laboratory

[CHEM 2323](#) Introductory Organic Chemistry I

[CHEM 2325](#) Introductory Organic Chemistry II

[PHYS 1301](#) College Physics I

[PHYS 1101](#) College Physics Laboratory I

or [PHYS 2125](#) Physics Laboratory I

[PHYS 1302](#) College Physics II

[PHYS 1102](#) College Physics Laboratory II

or [PHYS 2126](#) Physics Laboratory II

An additional 4 semester credit hours selected with your advisor from NSC or other fields. See advisor for additional guidance.

Elective Requirements: 12 semester credit hours of free electives.

At least 12 semester credit hours of lower- or upper-division courses of the student's choice. Students are encouraged to explore additional courses in Neuroscience as well as explore interests outside the field.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

Incoming freshmen must enroll and complete the requirement of [BBSU 1100](#).

Track 3: Industrial Neuroscience

Required coursework (16 semester credit hours)

[NSC 4360](#) Introduction to Entrepreneurship in Neuroscience

[NSC 4361](#) Pathway Into Industrial Neuroscience

[NSC 4193](#) Internship Preparation

Select 3 additional courses (9 semester credit hours) from this list:

[NSC 4391](#) Writing and Independent Study

[NSC 4394](#) Internship in Neuroscience

[NSC 4395](#) Internship in Neuroscience II

[NSC 4397](#) Thesis Research

[NSC 4V95](#) Externship in Neuroscience

[NSC 4V99](#) Independent Study

Additional Emphasis (6 semester credit hours)

Choose from one of the following two-course groups for additional emphasis:

Entrepreneurship

[ENTP 3301](#) Innovation and Entrepreneurship

[ENTP 4311](#) Entrepreneurial Strategy and Business Models

Healthcare Management

[HMG1 3301](#) Introduction to Healthcare Management

[HMG1 3310](#) Healthcare Regulatory Environment

Marketing

[MKT 3300](#) Principles of Marketing

[MKT 3330](#) Introduction to Professional Selling

Elective Requirements: (30 semester credit hours)

30 semester credit hours of free elective coursework, selected with your advisor, from NSC or other fields.

Electives are of lower- or upper-division courses of the student's choice. Students are encouraged to explore additional courses in Neuroscience as well as explore interests outside the field.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

Incoming freshmen must enroll and complete the requirement of [BBSU 1100](#).

Fast Track Baccalaureate/Master's Degrees

UT Dallas undergraduate students with strong academic records who intend to pursue a master's degree in Applied Cognition and Neuroscience at UT Dallas may consider an accelerated

undergraduate-graduate plan of study. If accepted into the program, students may take up to 15 semester credit hours of graduate courses that may be used to complete the baccalaureate degree and also satisfy requirements for the master's degree. Students must maintain a 3.000 grade point average and earn grades of B or better in graduate courses taken.

Students should apply for Fast Track admission in the semester they reach 90 semester credit hours. To qualify for application, undergraduate students must have completed at least 18 semester credit hours in major core courses at UT Dallas. To be eligible for Fast Track admission, students must have completed at least 90 semester credit hours toward a baccalaureate degree, completed a minimum of 36 hours of general education core curriculum classes, and meet program admission requirements. Apply to the Fast Track program through the Applied Cognition and Neuroscience Program Office. Students should consult with a graduate advisor regarding admissions criteria and plans of study.

Minors

Students must complete 18 semester credit hours including 9 required semester credit hours of foundation coursework and 9 semester credit hours of guided electives. At least 12 semester credit hours must be upper-division courses, of which at least 9 semester credit hours must have been completed at UT Dallas.

Minor in Neuroscience

18 semester credit hours

Students who are not majoring in Neuroscience may minor in Neuroscience. Students who take a minor will be expected to meet the normal prerequisites in courses making up the minor, and should maintain a minimum GPA of 2.000 on a 4.00 scale (C average).

Students should take 9 semester credit hours (3 courses) from the Neuroscience Core courses:

[NSC 3361](#) Introduction to Neuroscience

[NSC 4352](#) Cellular Neuroscience

[NSC 4354](#) Integrative Neuroscience

[NSC 4356](#) Neurophysiology

[NSC 4363](#) Neuropharmacology

[NSC 4366](#) Neuroanatomy

Plus 9 semester credit hours (3 courses) from one of the Neuroscience career tracks:

Medical Neuroscience (9 semester credit hours)

[NSC 4350](#) Medical Neuropathology

[NSC 4351](#) Medical Neuroscience

[NSC 4358](#) Neuroscience of Pain

[NSC 4362](#) Molecular Neuroscience

[NSC 4364](#) Journey into Medicine

[NSC 4372](#) Neuroimmunology

[NSC 4373](#) Sensory Neuroscience

[NSC 4382](#) Neurobiology of Emotion

Research Neuroscience (9 semester credit hours)

[NSC 4353](#) Neuroscience Laboratory Methods

[NSC 4357](#) Neurobiology of Learning and Memory

[NSC 4358](#) Neuroscience of Pain

[NSC 4362](#) Molecular Neuroscience

[NSC 4367](#) Developmental Neurobiology

[NSC 4371](#) Neural Plasticity

[NSC 4372](#) Neuroimmunology

[NSC 4373](#) Sensory Neuroscience

[NSC 4382](#) Neurobiology of Emotion

Industrial Neuroscience (9 semester credit hours)

[NSC 4360](#) Introduction to Entrepreneurship in Neuroscience

[NSC 4361](#) Pathway Into Industrial Neuroscience

[NSC 4391](#) Writing and Independent Study

[NSC 4193](#) Internship Preparation

[NSC 4394](#) Internship in Neuroscience

[NSC 4395](#) Internship in Neuroscience II

[NSC 4397](#) Thesis Research

[NSC 4V99](#) Independent Study

1. Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

2. A required preparatory course that also fulfills a Core Curriculum requirement. Eighteen (18) semester credit hours are counted in Core Curriculum.

3. May be repeated for credit, up to 9 semester credit hours.

4. May be repeated for credit, up to 6 semester credit hours.

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